

Experimental and numerical characterization of mechanical properties of carbon/jute fabric reinforced epoxy hybrid composites[†]

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Abstract

Natural fiber composites have great potential for reducing the product cost, lowering weight and enhancing renewability. Functionality and performance of natural fibers can be enhanced many folds using them together with synthetic fibers. Hybridization of carbon and low-cost natural jute fiber offers a sustainable hybrid composite having high modulus and mechanical strength. This study investigates flexural behavior of carbon/jute epoxy composites experimentally and numerically. Also, impact response is characterized through drop weight method. Study concludes that flexural strength decreases with increase in jute percentage. Simulation of flexural behavior diverges more than 10 % from experimental results. This anomaly is due to waviness of fiber resulting in heterogeneous property distribution in composites. Further, the fracto-graphic study revealed modes of failure. The drop weight impact tests reveal increased damage area with increase in jute percentage.

Keywords: CFRP; Drop weight test; Flexural test; Hybrid; Numerical simulation

1. Introduction

Fiber reinforced polymer composites are being utilized in various industries such as sports [1-3], automotive [4-6] and aerospace [7-9] because of their unique properties such as high strength to weight ratios, corrosion resistance and vibration absorption abilities as compared to metals. Laminated composites consist of several thin layers of fibers connected by a matrix. These laminated composites are generally reinforced with synthetic fibers like carbon and glass fibers. Lamination can be utilized to improve strength, stiffness, corrosion resistance, thermal insulation and moisture absorption. Synthetic fibers have high strength as compared to natural fibers but at the same time they are expensive. On the other hand natural fiber reinforcements stand out because of their low cost, biodegradability, renewability, low density and environmental friendly nature. Hybrid fiber composites are manufactured by using two or more types of reinforcements embedded in single matrix. Hybridization using natural and syn-

thetic fiber reinforcements is an efficient way of exploiting the use of natural fibers in structural applications [6]. Combination of carbon with a low cost natural fiber such as jute offers a more sustainable structure as compared to typical carbon fiber reinforced polymer (CFRP) structure. Jute adds its damping properties whereas carbon fibers bring their mechanical properties into the hybrid composite laminates. These hybrid composites can replace CFRP in structures prone to vibrations [10]. Strength of fiber reinforced composites depend on fiber properties, properties of matrix and the interfacial interaction between matrix and the fiber. In flexural loading exterior plies bear tensile and compressive stresses while neutral axis passes through the centroid where no stresses exists. It can be hypothesized that hybrid fiber reinforced composite structures will possess higher flexural [11] and impact strength when stiffer plies are placed at the exterior [12]. In the similar vein it can be expected that with the increase in the proportion of jute fiber layers, flexural and impact strength will decrease. Moreover the natural tendency of natural fibers to absorb moisture from the environment can be minimized by placing natural fiber layers as interior plies.

Citil employed non-linear three-dimensional finite element

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method to evaluate bending response of adhesively bonded curved lap joints. Experimental tests were also conducted to validate numerical results. Experimental and numerical solutions were found to be compatible [13]. Semsettin et al. performed non-linear analysis to predict response of bi-adhesive in double strap joint provided with an embedded patch under bending load. By means of bi-adhesive failure loads were found to show the increase ratio between 9 % and 56 % [14]. Vinsova and Urban conducted a numerical study on thick-walled CFRP composites to obtain flexural properties. Both unidirectional and woven reinforcements were utilized in composites fabrication. In order to obtain elastic properties of material, thin-walled CFRP composites were fabricated and tensile tests were performed to obtain tensile modulus strength and Poisson ratio. These elastic constants obtained were used in FEM simulations of thick-walled composites. The calculation error of simulation for unidirectional fiber reinforced composites was found to be 10 % while error for woven composites was higher [15].

Chaudhary et al. evaluated flexural strength of unfilled glass fiber reinforced polymer (GFRP) composite numerically and experimentally as per ASTM D790 standard. Error between FEM simulation and experimental results was found to be in range of 10 % [16]. Bajpai et al. performed flexural tests to evaluate flexural strength of different combinations of glass/jute epoxy composites. Composite having one glass and three jute plies displayed higher flexural strength [17]. Indra Reddy et al. conducted experiments on Jute, pineapple leaf and glass fiber epoxy composites according to ASTM D790 to find flexural strength. Increase in flexural strength was observed with increasing fiber volume fraction [18].

Gupta and Singh studied polylactic acid coated sisal fiber polyester composites experimentally to find flexural properties as per ASTM D790. Study demonstrated improvement in flexural strength using PLA coating on sisal fiber composites [19].

Ullah et al. determined bending response of woven GFRP composites through simulations using ABAQUS. Internal heterogeneity of woven composites resulted in slight divergence of FEM simulation results from those of experimental [20].

Soliman et al. determined on-axis and off-axis flexural strengths of carbon woven composites experimentally and numerically. FE simulations verified fiber dominance in on-axis flexural performance while matrix govern performance in off-axis flexural strength [21].

Gruber and Wartzack performed an anisotropic simulation to analyze three-point bending response of short fiber reinforced thermoplastic and obtained results were compared with experimental results. Integrative simulation for early design steps predicted absolute displacement values precisely [22].

Rafiquzzaman et al. conducted a numerical analysis of glass/jute reinforced epoxy hybrid composite laminates and then compared numerical results with experimental results. In this study, complete model was divided into different layers of composite plates to represent different mechanical properties for each layer. Isotropic type of material was assigned to com-

posite plates as an input in FEM simulation. Approximately, 15-20 % error was found between maximum stress values of simulation and experiment [23]. Dong et al. computed the flexural strengths of different stacking sequences of s-glass and carbon fiber reinforced epoxy hybrid laminates using classical laminate theory. Results obtained revealed validity of suggested model when compared with simulation results [24].

Different simulation options were also evaluated for GFRP rectangular composite beams consisting of two different adhesives polyurethane and epoxy [25].

Ramana and Ramprasad conducted experimental study to determine flexural strength of carbon-jute epoxy hybrid composite containing 22 % carbon, 23 % jute and 55 % epoxy. Research concluded that these hybrid composites can replace CFRP with small loss in flexural strength [26].

Abhishek et al. conducted probabilistic study of flexural strength of untreated jute reinforced polyester composites according to ASTM D7264. Maximum flexural strength was found in composites having 3.48 mm thickness and 15.62 % fiber volume fraction [27].

Braga and Magalhaes tested jute/glass epoxy hybrid composites experimentally to get flexural strength according to ASTM D790. Maximum flexural strength was found when glass plies were used at exterior [28].

Sezgin and Berkalp studied impact response of jute-carbon hybrid composites. Carbon fiber addition produced not only high tensile strength but also improved impact strength as compared to only jute composites [29]. Hung et al. performed a low velocity impact experimental analysis on different stacking sequences of glass/carbon reinforced polymer composites according to ASTM D7136. Samples containing surface carbon plies and glass plies in core presented high impact resistance in terms of local damage (delamination) while severe damage was found on samples with surface glass plies and core carbon plies [7].

Hongxiao et al. studied design optimization of CFRP stacking sequence for improving impact strength using a multi island genetic algorithms. Improved specimen was comprised of variety of angled plies as compared to base line laminate and presented 42.1 % less damage than base line laminate [30].

Sevkat et al. performed drop weight impact tests on woven hybrid composites using impactors of various geometries. glass/graphite/glass composite sustained higher impact loads while graphite/glass/graphite laminate displayed low delamination phenomenon [31].

Sarasini et al. conducted low velocity impact analysis on carbon/basalt epoxy hybrid laminates having two different combinations i.e. intercalated and sandwich-like. Post impact properties were characterized through flexural and inter laminar shear strength (ILSS) tests along with non-destructive tests: Acoustic emission and ultrasonic phased array. Higher impact energy absorption was found in intercalated sequence as compared to sandwich-like [32].

Gopinath et al. evaluated impact strength of jute fiber epoxy composites. Impact strength improved upon fiber treatment